

G14
Snowledge

Test Report

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Version history

Ver- sion	Date	Author	Description
0.1	13.11.2022	Sanni Onkalo	First draft
0.2	16.12.2022	Martin Pochat-Cottilloux	Add 1.1-1.4, 2.1-2.2,
0.3	16.12.2022	Micko Hakala	Add 2.3, 4.2, 4.3, 4.4, 4.5
0.4	16.12.2022	Sanni Onkalo	Modify 4 and finalize the document

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1 Introduction

1.1 Purpose and scope of document

Purpose of this document is to explain the tests that have been carried out during the project, the type of test and more generally to give stakeholders a complete overview of the whole testing process and the results obtained. This document covers unit testing, integration testing, system testing and acceptance testing.

1.2 Product and environment

Product is application called Snowledge, and it has been produced by Pallaksen Pöllöt. Application is developed to mobile platforms (Android, iOS) and for the browsers (web-application). Application offers current snow and weather conditions in Pallas area for people visiting it. Typical application user is skier or other explorer in Pallas area who wants to know about the environment and conditions in the area.

In the mobile application, there is GPS location feature that help's rescue personnel locate the victim of accident more quickly. If victim press help request button. From the mobile application, application will notify users nearby and personnel of Pallaksen Pöllöt about incident.

Project team focus mostly on the mobile application, so this document covers testing for it.

1.3 Project constraints related to testing

No constraints related to testing in this project.

1.4 Definitions, abbreviations, and acronyms

Exploratory testing	no test cases but story to follow
Test	testing of one requirement, functionality, or feature
Test case	everything required to one test
GPS	Global Positioning System
PM	Product Manager
GitLab	Version control, location of the source code
Flutter	Framework for building mobile applications
UI	User Interface
Codemagic	Tool for building CI/CD pipelines for mobile applications
MMT	Metrics Monitoring Tool

2 Testing process

The team followed **SOLID** principles when developing the application.

Single responsibility: "Every class should have only one responsibility"

The **Open-closed** principle: "Software entities should be open for extensions, but closed for modification"

The **Liscov** substitution principle: "Functions that use pointers or references to base classes must be able to use objects of derived classes without knowing it"

The **Interface segregation** principle: "Clients should not be forced to depend upon interfaces that they do not use"

The **Dependency inversion** principle: "Depend upon abstractions, not concretions."

2.1 General approach

There were no unit tests done in mobile application at the beginning of this project, so team initialized flutter tests. Before pushing the code, developer ran unit tests in his/her local development environment. Developers have also done integration testing for given tasks besides coding.

PMs did integration testing for the tasks before merging the code to the master branch.

Before publishing the application to the stores, team conducted system testing at their local development environments and iOS test environment TestFlight. After that, team published application to the Apple App Store and so the customer was able to do acceptance testing.

There are not a lot of users using the application currently, so acceptance testing can be done in production. It is the reason why the product was taken straight to production in the Google Play Store.

2.2 Testing roles

PMs worked as testing managers. PMs did system testing and they were part of integration testing while issues were on develop.

Developers did integration testing besides coding the task and is part of system testing.

Customer has joined the acceptance testing when the application was published on App Store.

2.3 Test schedule

Unit and widget testing was done continuously during the development. Plan was that system testing is starts during Sprint 5 and finished during Sprint 6. Customer can do acceptance testing during Sprint 6, starting form 8.12.2022.

Team was bit late with the system testing, and it started around mid of Sprint 5. Team was able to give product to the customer for the acceptance testing in 8.12.2022, so the plan for that was successful.

After 8.12, all the time went to the bug fixing and re-testing and final product were published to the stores 16.12.2022.

2.4 Test documentation

Flutter test gives feedback about unit testing. Statistics of unit and widget test cases have been documented in MMT during the project.

About system and acceptance testing, team had written user stories and test cases. While testing those cases, team has written documentation about testing results. All testing results are documented in appendix 2.

3 Testing Tools

CI/CD pipeline was executed in Codemagic as planned. The pipeline builds the application, runs automatic unit and widget tests, and executes static code analysis. Pipeline is triggered once code is pushed to master branch. Possible failed tests will result in a new issue in Gitlab.

The team implemented 2 unit tests to the automated pipeline. The automated pipeline acted mostly as a build test, which ensured that each version of the master branch builds successfully.

Statistics of unit and widget test cases were documented in MMT during the project.

4 Test cases and results

4.1 Test results

System testing passed for version 2.0.3. The results were as follows:

- Test cases: 6
- Passed cases: 6
- Open issues: 1

More detailed results can be found in appendix 2.

Acceptance testing is still in process at the time of writing this.

4.2 Unit testing

Two test cases were written with flutter test to test the UI. One of those two unit test did not pass, and team were not able to locate the problem. The UI works perfectly so probably there is some issue with the test.

Unit testing results were as follows:

- Test cases: 2
- Passed: 1
- Ignored: 1

4.3 Integration testing

Developers did integration testing for the task they are working on. PM's did integration testing if needed for the task they are code reviewing. All the issues that were merged to the master were passed by the integration test.

4.4 System testing

System testing was done by Project team at the end of the project. Selected method for system testing is use case testing. Use cases are described appendix 1 and System testing results are documented in appendix 2 and 3.

4.5 Acceptance testing

Customer does acceptance testing utilizing use cases described in appendix 1. Acceptance testing is done at the end of the project, after Project team has done system testing.

Customer was happy with the outcome but were not able to test all the cases before this report were delivered.

4.6 Exploratory testing

Team and customer utilized exploratory testing to test the user flow for the app. One requirement for the project was to have an easy-to-use user interface, and for that requirement exploratory testing was the most suitable approach. Exploratory testing method revealed functional and visual bugs, which the customer reported to the team via email. All bugs on version 2.0.2, which were found and reported, were fixed to version 2.0.3. This document does not cover those findings in more detail.

APPENDIX

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|--------------------------------|----------------------------|
| 1. Use cases | Snowledge-UseCases.pdf |
| 2. Test results, version 2.0.3 | 2022-12-16-TestResults.pdf |
| 3. Test results, version 2.0.2 | 2022-12-15-TestResults.pdf |